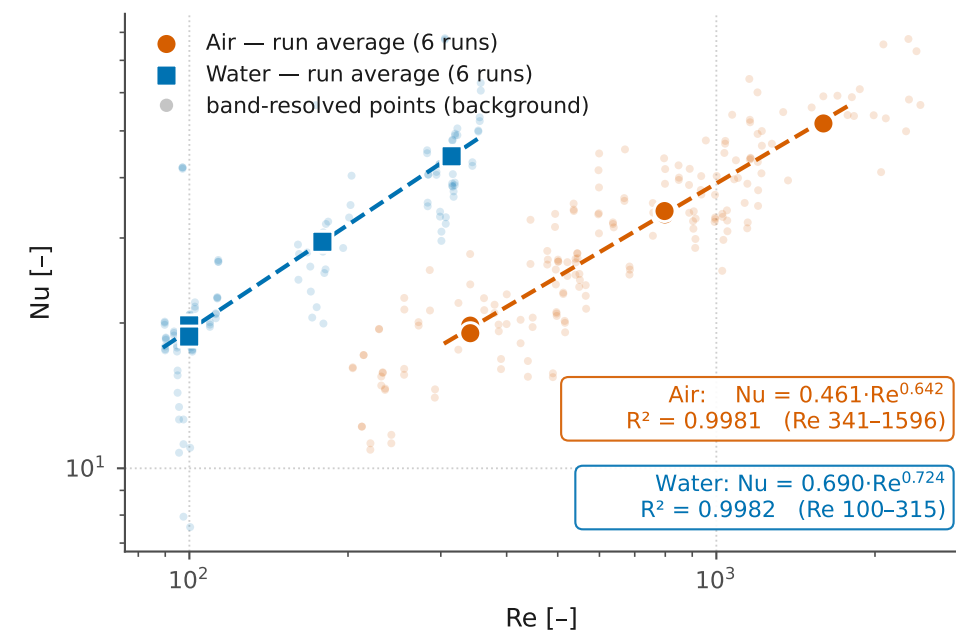
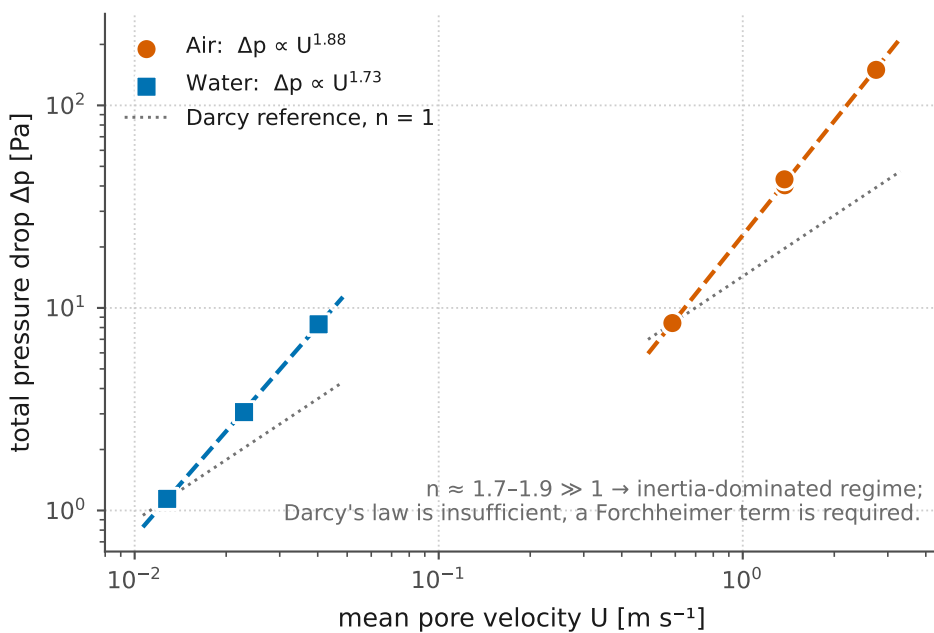


Band-resolved analysis of a TPMS gyroid heat exchanger: from six flow regimes to a design correlation set

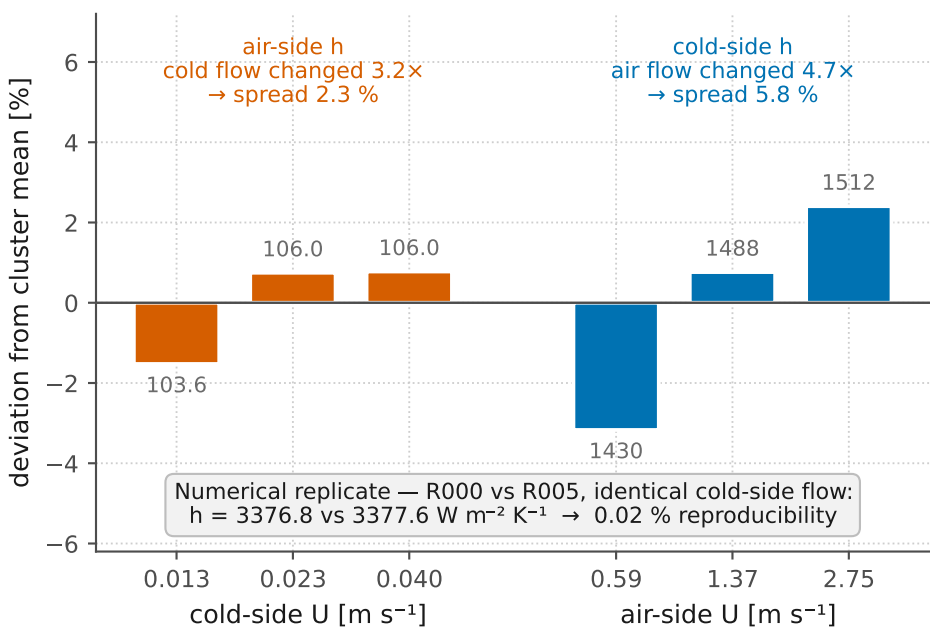
(a) Design correlation $Nu(Re)$ — both streams



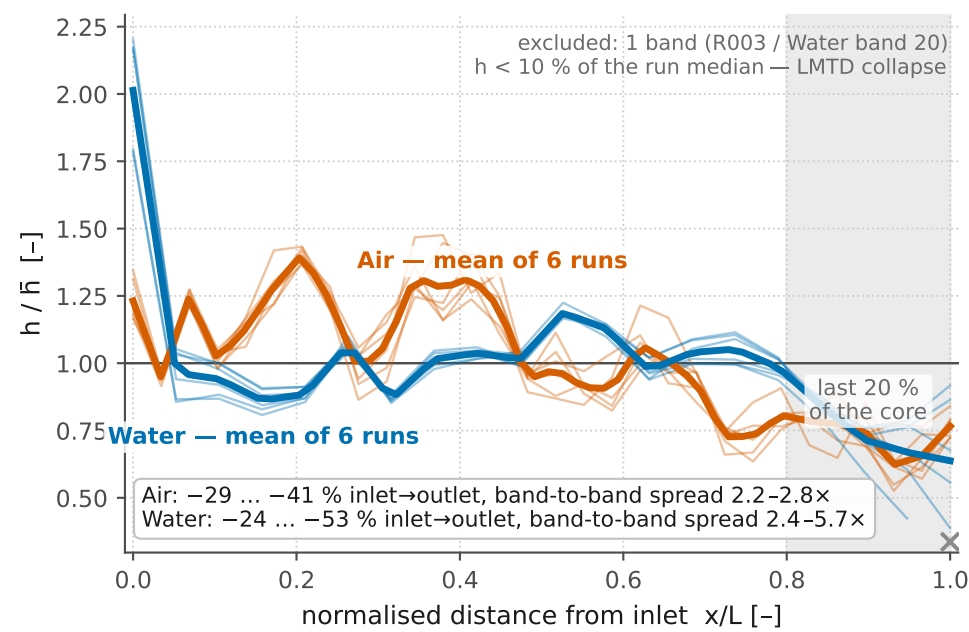
(b) Pressure-drop law $\Delta p(U)$ — inertial despite laminar flow



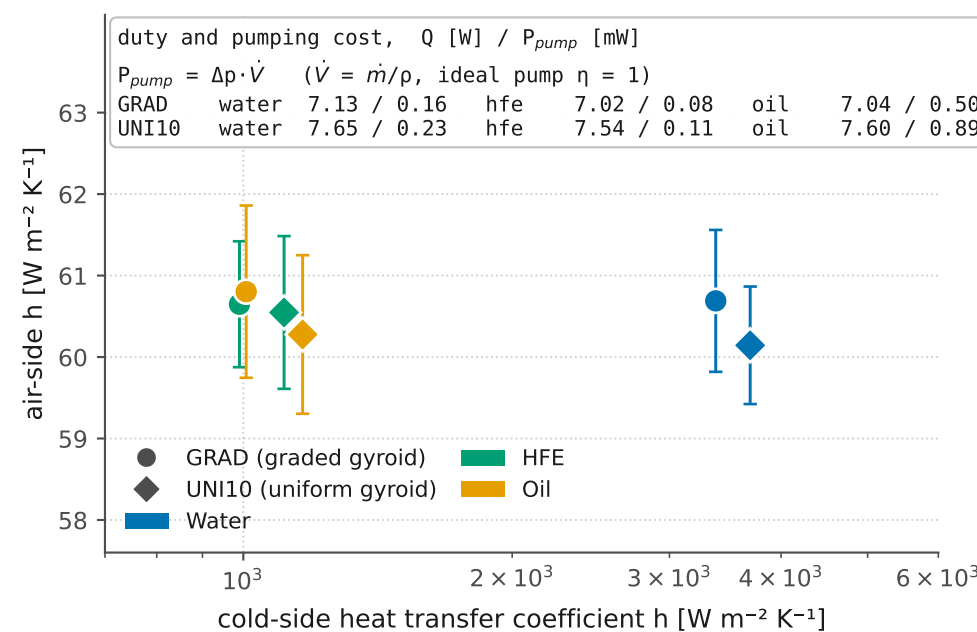
(c) The two sides are decoupled — and the solver repeats itself



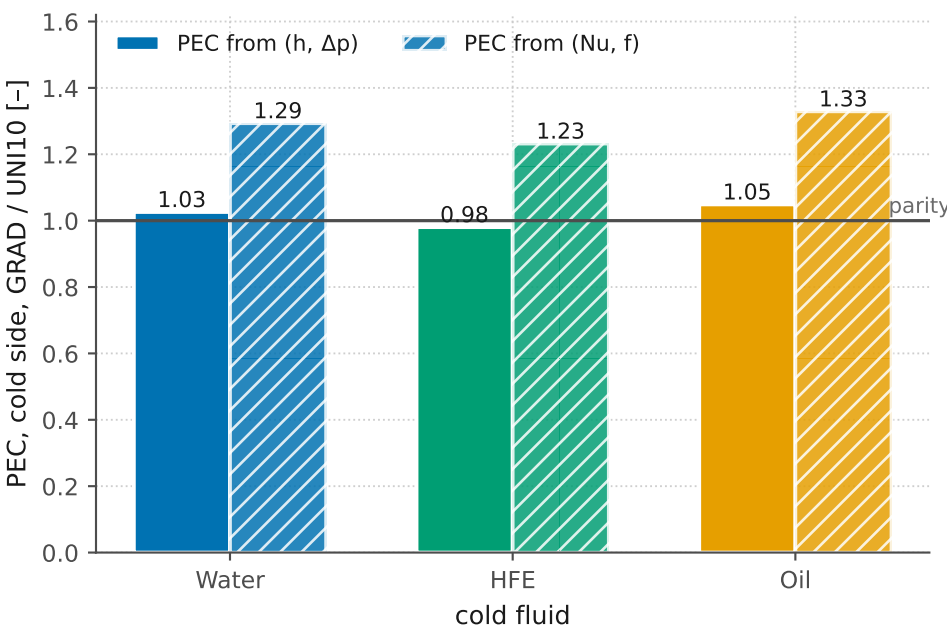
(d) The mean hides a 2-6× band-to-band variation



(e) Cold-fluid choice sets the pumping cost, not the duty



(f) Verdict at equal pumping power — and the validation behind it



cold-side h varies 3.7× → duty stays at $Q = 7.02-7.65\ W$;
on UNI10 oil costs 3.9× the pumping power of water for the same duty

$PEC = (X_G/X_U)/(Y_G/Y_U)^{1/3}$, $X, Y = (h, \Delta p)$ or (Nu, f) , $G/U = GRAD/UNI10$; the 1/3 exponent → equal pumping power
Validation (transient window 18–19 s, 50 time steps)
energy balance $|Q_{air} - Q_{cold}|$: 0.33–0.67 % mean, ≤ 1.5 % worst step
temporal scatter of h : air 1.2–1.7 %, cold liquid 0.02–1.14 %